

Kenny Mustapha

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EDUCATION

Simon Fraser University

Bachelor of Science in Computer Science

Burnaby, BC

Expected May 2028

EXPERIENCE

Student Software Engineer

May 2026 – Aug 2026

Develop for Good

Remote

- Built a multi-vendor marketplace using Next.js and PostgreSQL for Ikhaya Lethu Care for the Elderly, enabling community vendors to sell products online and generate sustainable funding for elderly care programs.
- Collaborated directly with the client to translate requirements into shipped marketplace features; added validation across vendor and storefront workflows to improve data integrity and reliability.

Software Engineering Intern

Jan 2026 – Apr 2026

Lillup

Remote

- Parallelized Python text-to-speech benchmarking pipelines to automate latency and quality comparisons across model variants, cutting experiment turnaround time by approximately 40%.
- Built a precision-aware validation harness comparing FP16 and FP32 execution across model layers, reducing manual validation effort by approximately 60% while preserving output quality.
- Instrumented latency, memory use, and fidelity across model configurations to support production deployment decisions.

Machine Learning Research Assistant

Jan 2026 – Apr 2026

SFU School of Computing Science

Burnaby, BC

- Built Python computer vision pipelines with YOLO, MediaPipe, and OpenCV to evaluate pose landmarks, object detections, and spatial measurements across live and controlled experiments.
- Implemented camera calibration and monocular depth-estimation workflows, achieving 0.03 px reprojection error and validating physical measurements across desktop and Raspberry Pi deployments.
- Exposed experiment outputs through FastAPI as structured JSON, improving reproducibility and downstream integration.

Software Engineering Intern

Sep 2025 – Dec 2025

Deepiri

Remote

- Profiled PostgreSQL endpoints under concurrent load and introduced Redis caching and connection pooling, reducing repeated database queries by approximately 30% across cacheable API routes.
- Built Python observability pipelines for latency, cache hit rate, and connection pool utilization, helping engineers identify backend bottlenecks during production incidents.
- Worked with engineers to prioritize backend optimizations from profiling results, improving API reliability and performance under traffic spikes.

PROJECTS

Semantic Git Blame | *Python, FastAPI, PostgreSQL, pgvector*

- Built a developer tool that ingests commit history, pull-request descriptions, and linked issues into PostgreSQL and pgvector for semantic retrieval.
- Mapped selected code to relevant commits and engineering discussions, ranked retrieved context, and generated commit-linked explanations with an LLM.
- Exposed the retrieval system through a FastAPI service and CLI, returning sourced explanations instead of raw blame output.

LLM API Gateway | *Go, PostgreSQL, Redis, Docker*

- Built a self-hosted Go API gateway that provides one interface for OpenAI, Anthropic, and Cohere with model routing, rate limiting, and live cost tracking.
- Implemented a Redis-backed token bucket for per-key quotas and a PostgreSQL ledger for model, token count, latency, and request-cost logging.
- Added provider fallback that retries failed requests against secondary providers without requiring client-side changes.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Go, Java, SQL, C++

Frameworks & Libraries: FastAPI, Node.js, Express.js, Next.js, React, React Native

Databases & Tools: PostgreSQL, Redis, MongoDB, Firebase, Prisma, Git, Docker

AI/ML: PyTorch, scikit-learn, OpenCV, YOLO, MediaPipe, NumPy, Pandas, pgvector